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Corporate Climate Risk: Measurements and Responses

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1 Big picture

- **Question.** How can researchers measure firm-level exposure to climate risk, and how do capital markets and firms respond to that exposure?
- **Measurement contribution.** The paper constructs quarterly, firm-level climate risk measures using textual analysis of earnings call transcripts. It separately measures acute physical risk, chronic physical risk, and transition risk.
- **New response dimension.** Within transition risk, the paper distinguishes discussions with proactive language from discussions without proactive language. This creates a measure of whether firms appear to actively manage transition risk.
- **Sample.** The main sample uses Thomson Reuters StreetEvents earnings calls for U.S. public firms from 2002 through the first half of 2018, matched to Compustat, CRSP, ESG scores, natural disaster data, EPA emissions, RavenPack news, SEC filings, and patent data.
- **Scale.** The baseline sample contains 4,719 public firms and 139,959 firm-quarter observations after excluding financial firms and firms headquartered outside the continental United States.
- **Main validation result.** The text-based physical risk measures line up with natural disaster exposure. The transition risk measure lines up with MSCI's Climate Change Index and direct CO₂ intensity.
- **Main asset-pricing result.** High transition risk is associated with lower Tobin's q , especially after 2010 when climate attention and regulation increased.
- **Key nuance.** The valuation discount is driven by the nonproactive component of transition risk. Firms that discuss transition risk with proactive language are not penalized in the same way.
- **Robustness result.** The earnings-call transition measure continues to matter in horse-race regressions against SEC filing measures, news-based measures, MSCI climate scores, and Sautner et al.-style climate exposure measures.
- **Real response result.** Proactive firms increase capital expenditures and green innovation when transition risk rises, while nonproactive firms reduce R&D and employment.
- **Economic takeaway.** Climate risk is not a single scalar. The market appears to distinguish between firms that merely face transition risk and firms that actively respond to it.

2 Data and measurement

2.1 Earnings calls as the main textual source

- The paper uses transcripts of quarterly earnings conference calls from Thomson Reuters StreetEvents.
- Earnings calls contain management presentations and analyst/investor Q&A, making them less purely scripted than many regulatory filings.
- The transcript data allow climate risk to be measured at the firm-quarter level.
- The sample runs from January 2002 through the first half of 2018.
- The paper extracts full call text and identifiers such as firm names, tickers, CUSIP numbers, dates, and times.

Interpretation: Earnings calls are useful because they capture forward-looking management discussion and analyst questions. This makes them better suited than static emissions data for measuring changing firm exposure and management attitudes.

2.2 Financial variables and firm outcomes

- **Valuation outcome:** Tobin’s q from Compustat.
- **Investment outcome:** capital expenditures scaled by lagged assets.
- **Innovation outcomes:** R&D expenditures, green patent indicator, and green patents as a share of total patents.
- **Employment outcome:** log annual employment.
- **Controls:** log assets, CapEx, PPE, book leverage, ROA, number of analysts, institutional ownership, institutional HHI, and energy-price exposure.
- Financial firms are excluded because insurance and financial intermediation can make climate-risk discussions qualitatively different.

Interpretation: The paper connects climate language to both market valuation and real firm behavior. This lets the authors ask whether climate risk is priced and whether firms respond in investment, innovation, and labor demand.

2.3 Constructing climate dictionaries

- The authors construct three separate dictionaries: acute physical risk, chronic physical risk, and transition risk.
- Physical risk is split into acute events such as hurricanes, floods, droughts, and wildfires, and chronic conditions such as abnormal weather, temperature, precipitation, and seasonal changes.
- Transition risk captures business exposure to policies, regulation, technology shifts, and changing preferences in the move toward a low-carbon economy.
- The dictionaries contain both unigrams and bigrams to reduce false positives and false negatives.
- The final dictionaries contain 37 unigrams and 1,649 bigrams: 21 unigrams and 350 bigrams for acute physical risk, 16 unigrams and 977 bigrams for chronic physical risk, and 322 bigrams for transition risk.

Interpretation: The dictionary approach is deliberately transparent. The authors prefer it to black-box machine learning because climate language is subtle and context-dependent.

2.4 Firm-level climate risk measures

For a firm i and quarter t , the text-based climate risk measure is conceptually

$$ClimateRisk_{i,t}^c = 10^4 \times \frac{\sum_{b \in \mathcal{D}_c} Freq_{i,t,b}}{Length_{i,t}}, \quad (1)$$

where $\mathcal{D}_c$ is the dictionary for climate category c , $Freq_{i,t,b}$ is the frequency of keyword or bigram b , and $Length_{i,t}$ is transcript length.

For physical risk, keyword mentions must occur near a risk synonym:

$$PhysicalRisk_{i,t}^c = 10^4 \times \frac{\sum_{b \in \mathcal{D}_c} Freq_{i,t,b}^{near \ risk}}{Length_{i,t}}, \quad c \in \{acute, chronic\}. \quad (2)$$

For transition risk, proximity to risk synonyms is not required:

$$TransitionRisk_{i,t} = 10^4 \times \frac{\sum_{b \in \mathcal{D}_{transition}} Freq_{i,t,b}}{Length_{i,t}}. \quad (3)$$

Interpretation: The proximity restriction for physical risk prevents the measure from counting generic weather references as firm risk. Transition risk is broader because transition discussions often concern policies, technologies, or business strategy rather than immediate risk language.

2.5 Proactive and nonproactive transition risk

- The paper identifies verbs near transition-risk keywords that signal proactive responses, such as achieve, build, develop, improve, invest, prepare, reduce, replace, respond, restructure, and spend.
- Transition risk is decomposed into:

$$TransitionRisk_{i,t} = TransitionRisk/Proactive_{i,t} + TransitionRisk/Nonproactive_{i,t}. \quad (4)$$

- The proactive component captures transition-risk discussion near action-oriented verbs.
- The nonproactive component captures transition-risk discussion without that proactive language.

Interpretation: This decomposition is central to the paper. The market’s response depends less on whether climate risk is mentioned and more on whether the firm appears to be actively managing it.

2.6 External validation data

- **Natural disasters:** county-level SHELDUS data matched to firm headquarters locations.
- **ESG/climate scores:** MSCI Climate Change Index is the main external benchmark; RepRisk and Refinitiv environmental scores are supplemental benchmarks.
- **Emissions:** EPA Greenhouse Gas Reporting Program plant-level CO₂ emissions matched to Compustat firms.
- **Alternative text sources:** 10-K/10-Q MD&A sections, 10-K/10-Q risk factor sections, and RavenPack news.
- **Innovation data:** Global Corporate Patent data, including green patents.

Interpretation: The validation logic is that a useful measure should correlate with external information in the right places: physical risk with natural disasters, transition risk with climate-management scores and emissions, and proactive risk with real green responses.

3 Empirical specifications and interpretation

3.1 Validation of physical risk measures

The paper relates physical risk measures to local natural disaster exposure:

$$ClimateRisk_{i,t+1} = \sum_{p=0}^3 \beta_p Z_{c,t-p} + \gamma' X_{i,t-1} + \zeta_{i,t} + \varepsilon_{i,t}, \quad (5)$$

where $Z_{c,t-p}$ is whether county c has a natural disaster in lagged period $t - p$.

Interpretation: A positive coefficient means that firms located in counties with recent disasters discuss more acute or chronic physical climate risk in the following earnings call.

3.2 Validation of transition risk measures

The paper validates transition risk using MSCI CCI and EPA CO₂ intensity:

$$ExternalMeasure_{i,t+h} = \alpha + \beta TransitionRisk_{i,t} + \gamma' X_{i,t-1} + FE + \varepsilon_{i,t+h}. \quad (6)$$

Interpretation: If transition risk is a useful exposure measure, it should be positively associated with climate-change management scores and emissions intensity.

3.3 Pricing transition risk

The main valuation regression is

$$TobinQ_{i,t+h} = \alpha + \beta TransitionRisk_{i,t} + \delta EnergyPriceExposure_{i,t} + \theta ActionIndex_{i,t} + \gamma' X_{i,t-1} + FE + \varepsilon_{i,t+h}. \quad (7)$$

The decomposed version replaces $TransitionRisk_{i,t}$ with proactive and nonproactive components:

$$TobinQ_{i,t+h} = \alpha + \beta_N Nonproactive_{i,t} + \beta_P Proactive_{i,t} + \delta EPE_{i,t} + \theta ActionIndex_{i,t} + \gamma' X_{i,t-1} + FE + \varepsilon_{i,t+h}. \quad (8)$$

Interpretation: The coefficients β_N and β_P answer the key economic question: does the market discount all transition risk, or only transition risk that is not accompanied by proactive management language?

3.4 Within-firm pricing specification

To examine changes within firms, the paper estimates

$$\Delta TobinQ_{i,t+h} = \alpha + \beta \Delta TransitionRisk_{i,t} + \gamma' X_{i,t-1} + FirmFE + Industry \times TimeFE + \varepsilon_{i,t+h}. \quad (9)$$

Interpretation: This design asks whether changes in a firm's own climate-risk discussion predict changes in its valuation, reducing concern that the result is purely cross-sectional.

3.5 Firm response specifications

For real outcomes, the generic specification is

$$Y_{i,t+h} = \alpha + \beta_N Nonproactive_{i,t} + \beta_P Proactive_{i,t} + \delta EPE_{i,t} + \theta ActionIndex_{i,t} + \gamma' X_{i,t-1} + FE + \varepsilon_{i,t+h}, \quad (10)$$

where Y is CapEx, R&D, green patents, green patent ratio, or employment.

Interpretation: These regressions distinguish between exposure and response. A firm may face transition risk, but whether it responds proactively determines whether investment, green innovation, and employment adjust.

4 Identification and empirical design

4.1 What the design does well

- Firm-quarter text gives high-frequency variation in risk exposure.

- Industry-by-time fixed effects absorb common sector-level trends in climate policy and investor attention.
- Firm attributes control for firm size, leverage, investment intensity, analyst coverage, institutional ownership, and profitability.
- The proactive/nonproactive split offers a mechanism test: the valuation discount should be concentrated among firms that do not appear to respond.
- Horse-race regressions show that the earnings-call measure contains information not captured by filings, news, MSCI, or alternative climate dictionaries.
- Within-firm change specifications reduce reliance on static cross-sectional differences.

4.2 Main identifying assumption

The pricing interpretation requires that, conditional on controls and fixed effects, firms with higher transition-risk discussion are not simply firms with some other unobserved trait that mechanically lowers Tobin's q .

Interpretation: The paper cannot fully randomize climate risk exposure. Instead, it builds credibility through validation tests, rich controls, fixed effects, alternative measures, horse races, and within-firm changes.

4.3 Potential limitations

- Earnings calls are voluntary disclosure environments, so managers may strategically disclose or avoid climate topics.
- Public firms with earnings calls are not representative of all firms.
- Text measures capture discussion of risk, not the full structural exposure of assets, operations, or supply chains.
- The proactive language measure captures textual proactiveness; it does not prove that every stated action is effective.
- Capital-market pricing results are predictive associations rather than clean causal estimates.

Interpretation: The strongest reading is that the paper provides a validated measurement framework and documents robust correlations between measured climate risk, market valuation, and firm responses.

5 Tables and figures

5.1 Table 1 and Table 2: What is measured and which words drive the measures

Read:

- Table 1 shows that all climate-risk variables are sparse: the median and 75th percentile are zero for acute, chronic, transition, proactive, and nonproactive measures.
- Transition risk is much larger than physical risk on average: mean transition risk is 3.38 with a standard deviation of 13.17.
- The nonproactive component dominates the proactive component: mean nonproactive transition risk is 3.05, while mean proactive transition risk is 0.32.
- The baseline sample has 139,959 firm-quarter observations for the text measures.
- Table 2 shows intuitive keywords: hurricanes, storms, drought, flooding, and wildfire for acute risk; weather, temperatures, snow, and precipitation for chronic risk; and energy efficiency, renewable energy, solar, clean energy, alternative energy, and carbon for transition risk.

Economic message: The text measures are sparse but informative. The climate terms are economically meaningful and the high transition-risk frequency reflects the broad relevance of energy technology and policy.

Table 1
Summary statistics

Variable	N	Mean	SD	Min	P25	P50	P75	Max
Firm-level measures constructed from earnings calls								
Acute Climate Risk	139,959	0.06	0.61	0.00	0.00	0.00	0.00	11.75
Chronic Climate Risk	139,959	0.20	1.26	0.00	0.00	0.00	0.00	17.72
Transition Climate Risk	139,959	3.38	13.17	0.00	0.00	0.00	0.00	186.59
Transition Risk/Proactive	139,959	0.32	1.70	0.00	0.00	0.00	0.00	22.40
Transition Risk/Nonproactive	139,959	3.05	12.10	0.00	0.00	0.00	0.00	174.03
Energy Price Exposure	139,959	0.00	0.01	0.00	0.00	0.00	0.00	0.07
Action Index	139,959	0.02	0.01	0.01	0.02	0.02	0.02	0.04
Other firm-level data								
Tobin's q	130,450	2.03	1.50	0.46	1.16	1.56	2.32	14.82
CapEx	136,121	2.89	3.73	0.00	0.65	1.60	3.54	21.03
R&D	138,169	1.35	2.62	0.00	0.00	0.00	1.72	14.23
log(Asset)	138,208	6.84	1.92	-1.62	5.54	6.83	8.13	13.65
PPE	134,158	0.25	0.24	0.00	0.07	0.16	0.37	0.89
Book Leverage	130,244	0.24	0.23	0.00	0.03	0.21	0.37	1.01
log(No_Analysts)	139,959	1.83	0.89	0.00	1.39	1.95	2.48	3.93
Institution %	135,383	0.67	0.27	0.00	0.51	0.75	0.89	1.00
Institution HHI	134,985	0.10	0.13	0.01	0.04	0.05	0.09	1.00
ROA	136,881	0.06	0.23	-0.96	0.03	0.11	0.17	0.46
log(Employment) (annual)	38,917	1.45	1.29	0.00	0.34	1.12	2.24	7.74
External data								
Disaster dummy	139,959	0.05	0.22	0.00	0.00	0.00	0.00	1.00
CO ₂ Intensity (annual)	2,774	4.12	7.97	0.00	0.23	0.97	4.08	52.93
Green patents (annual)	39,505	0.08	0.27	0.00	0.00	0.00	0.00	1.00
Green patents ratio (annual)	12,664	0.04	0.14	0.00	0.00	0.00	0.00	1.00
MSCI CCI	17,304	56.44	66.62	0.00	0.00	33.00	94.90	594.00
RepRisk Environmental Score	40,925	2.15	4.89	0.00	0.00	0.00	0.00	31.51
Refinitiv Environmental Score	49,351	47.39	21.70	6.51	29.97	43.20	64.19	97.82
Firm-level measures constructed from alternative data								
Transition Risk MDA	108,714	2.82	8.54	0.00	0.00	0.00	1.39	95.20
Transition Risk RF	89,999	2.16	8.96	0.00	0.00	0.00	0.00	108.06
Transition Risk News	139,959	0.01	0.06	0.00	0.00	0.00	0.00	0.67

This table reports the summary statistics of all variables used in the regression analysis. All variables are at the firm-quarter level, except that *log(Employment)*, *CO₂ Intensity* and green-patent-related variables are at the firm-year level. All the climate risk variables, including the acute, chronic, and transition climate risks are explained in Section 2 and the statistics are summarized after winsorization, but before standardization. Table A.1 in the appendix contains detailed definitions of all variables.

Table 1: Table 1: Summary statistics

Table 2
Top climate-related keywords

Acute risk			Chronic risk			Transition climate risk		
Bigram/ Unigram	Freq	$\frac{\text{freq}}{P_T} \times 10^4$	Bigram/ Unigram	Freq	$\frac{\text{freq}}{P_T} \times 10^4$	Bigram/ Unigram	Freq	$\frac{\text{freq}}{P_T} \times 10^4$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
hurricane	1560	6371.9	weather	6154	26342.7	energy efficiency	7738	32512.0
hurricanes	552	2243.5	temperatures	122	506.0	renewable energy	6663	28064.3
storms	409	1652.7	the snow	75	299.4	the solar	6623	28819.0
drought	294	1177.2	high water	72	296.2	clean energy	5117	21372.2
flood	185	728.7	heating season	49	200.4	alternative energy	4160	18367.0
the flood	108	440.6	precipitation	46	252.1	superior energy	3354	12482.7
wildfire	110	356.4	the ice	40	237.1	higher energy	2806	11273.8
wildfires	75	331.8	the ice	37	216.7	new energy	2503	10878.1
storm losses	54	201.6	mild winter	48	188.8	the renewable	2399	10564.8
severe winter	30	155.4	snowfall	42	186.8	the ecosystem	2590	10036.0
storm related	33	134.0	rainfall	42	175.4	energy management	2156	8801.2
the floods	24	102.0	degree days	34	173.9	energy efficient	2171	8459.6
storm activity	25	100.8	winter conditions	36	170.7	the carbon	2243	8414.0
storm costs	21	86.8	warm winter	36	161.0	green energy	2224	8303.4
water flood	22	82.4	cold winter	34	138.0	wind energy	1893	7817.5
polar vortex	22	76.8	hot summer	33	126.4	the climate	1926	7504.6
storm season	14	69.7	unseasonably warm	24	101.1	fuel efficiency	1874	6730.5
storm damage	10	64.4	the fog	26	107.4	shale gas	1655	6350.9
droughts	14	57.4	harsh winter	27	103.5	lower energy	1553	6290.3
extreme cold	13	55.3	unseasonably cold	19	99.6	fuel efficient	1592	5925.9
extremely cold	10	40.0	the clouds	23	96.7	energy technologies	1643	5883.5
storm cost	11	39.0	the warmest	12	74.5	wind farm	1444	5836.2
the volcano	11	38.3	early winter	13	74.1	alternative fuel	1301	5776.1
			hailstorms	13	72.3	wind farm	1283	5686.7
			cold season	17	70.9	fuel economy	1586	5487.9
			the rain	16	64.7	the oil	1479	5476.3
			wind hail	11	63.2	solar cell	1170	5475.9
			the winds	17	62.8	gas drilling	1286	4947.6
						energy future	1214	4715.9
						storm projects	1076	4467.6

This table lists the top-30 unigrams or bigrams in each category of *ClimateRisk_{it}* measures, ranked by *freq_{it}*. To calculate the *freq_{it}* for acute and chronic climate risk measures, we first identify the frequency of mentions of individual unigrams and bigrams in proximity to risk synonyms (*freq_{it}*, *p_{it}*). We then divide the frequency by the length of the transcript (*P_{it}*), multiply the quotient by 10⁴, and sum the resultant values across all transcripts in our sample. The calculation of *freq_{it}* in the case of transition climate risk is the same except that we do not require the mention of the unigrams and bigrams to be in the proximity of risk synonyms, which leads to higher *freq_{it}* and *freq_{it}* for that specific category.

Table 2: Table 2: Top climate-related keywords

5.2 Figures 1 and 2: Time-series and industry variation

Read:

- Figure 1, panel A shows spikes in acute physical risk around major hurricane events and related weather shocks.
- Figure 1, panel B shows transition risk rising strongly through the late 2000s and then remaining elevated relative to the early sample.

- Figure 1, panel C shows that both proactive and nonproactive transition-risk components rise, but nonproactive transition risk is typically larger.
- Figure 2 shows strong industry heterogeneity. Utilities are especially exposed to transition risk, while physical risk is higher in industries such as agriculture, mining, utilities, construction, and arts/recreation.

Economic message: Climate risk exposure is both time-varying and industry-dependent. The time-series pattern supports the idea that climate discussions react to real events and evolving transition pressure.

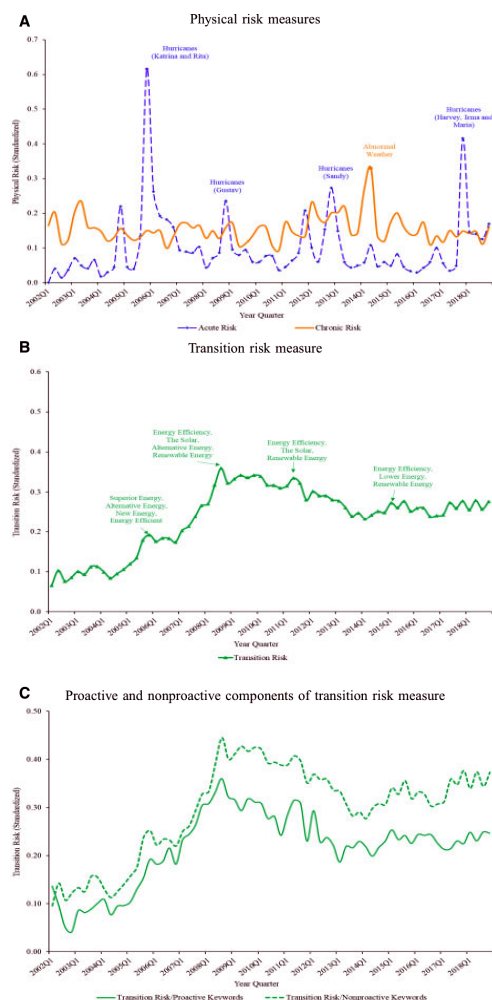


Figure 1: Figure 1: Average firm-level climate risk measures over time

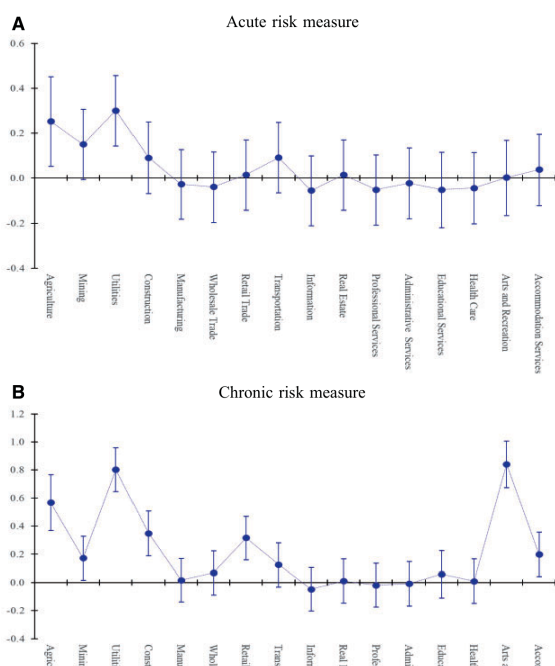


Figure 2: Figure 2, panels A-B: industry variation in acute and chronic risk

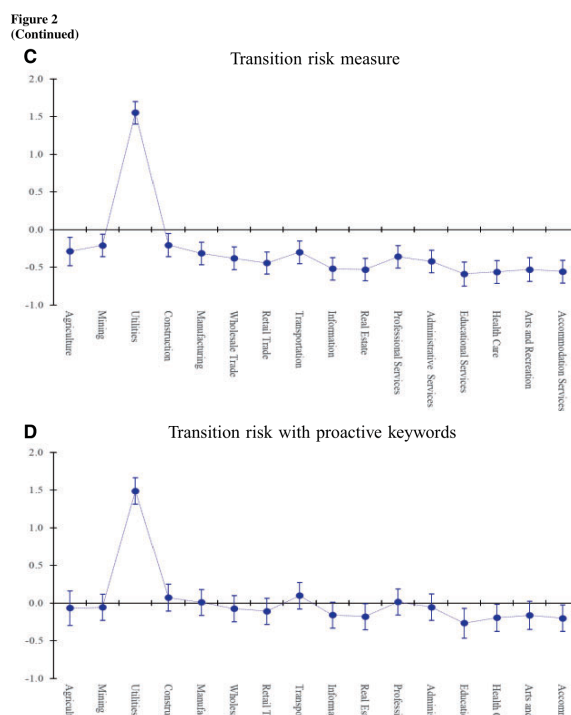


Figure 3: Figure 2, panels C-D: industry variation in transition and proactive transition risk

5.3 Table 3, Table 4, and Figure 3: Firm-level examples and validation

Read:

- Table 3 gives transcript excerpts with the highest measured climate risks. The acute examples include wildfire and hurricane discussions; chronic examples include abnormal weather and temperatures; transition examples include clean energy, renewable energy, emission reduction, and efficiency language.
- Table 4 validates physical risk by showing that natural disasters predict higher acute and chronic climate-risk discussions.
- Table 4 also validates transition risk by showing positive associations with MSCI CCI and CO₂ intensity.
- Figure 3 visually shows positive binned relationships between transition risk and MSCI CCI, and between transition risk and direct CO₂ intensity.

Economic message: The validation exercises show that the text measure is not just generic climate talk. It lines up with external measures of physical events, climate scores, and emissions.

Table 3
Excerpts in transcripts with highest climate risks

Firm	Date	Climate risk	Value	Keywords	Text surrounding the keywords
Edison International	Oct. 30, 2018	Acute risk	40.00	Wildfire; Uncertainty	We also have the flexibility at these entities to obtain both short and long-term debt while we continue to evaluate options as we work through uncertainty around the wildfire liability and cost recovery.
PG&E Corp	Nov. 5, 2018	Acute risk	39.85	Wildfire; Risks	Our expanded Community Wildfire Safety Program was established after the 2017 wildfires to implement additional precautionary measures intended to reduce or further reduce wildfire risks.
Parist Transportation	Nov. 30, 2017	Acute risk	35.63	Hurricane; Unpredictable	Hurricane Irma more directly impacted our operations as the state of Florida shut down for 2 or 3 days. This type of business is generally less productive with long lines, unpredictable traffic patterns and other negative occurrences leading to inefficient utilization of our equipment.
Softly Hotels Inc.	Nov. 8, 2016	Acute risk	32.40	Hurricane; Unsure	Heading into that market's high winter season we are unsure what the effects may be. The impact of hurricane Matthew on our portfolio in early October was significant.
Talos Petroleum LLC	Nov. 5, 2008	Acute risk	29.00	Storm; Risk	We're also actively engaged in a program of accelerated idle well abandonment to mitigate the ongoing risk of future storms.
Spert Child Inc	Nov. 15, 2018	Chronic risk	77.72	Weather; Variability	While the heating season presented some extreme weather variability, average temperatures across our service territories were 8% cooler than the prior year.
Partners	Feb. 6, 2013	Chronic risk	63.22	Unseasonably warm; Uncertainty	Unseasonably warm and dry weather coming on top of a bad winter sports season last year, combined with our customers' general economic uncertainty along with our desire to be less promotional, all contributed to the slight decline in comparable store sales.
Macys Inc	Feb. 18, 2016	Chronic risk	61.79	Precipitation; Chance	According to the National Oceanic Atmospheric Administration, in March through May, we are looking at about a 33% to 40% chance of above-normal precipitation in the southern portion of our service area and normal precipitation levels in the northern portion.
CH Energy Group Inc	Apr. 24, 2002	Chronic risk	52.52	Weather; Risk	A certain amount of variation from normal, either above or below normal degree days was a variation or risk that we estimated. Then there was a wider range where we would be compensated if weather were warmer than normal.
Southern Company Gas	Oct. 30, 2013	Chronic risk	51.63	Weather; Unpredictable	Given where you see the rates today, when you're coming up for the 2014 expiration, do you expect - doesn't seem to have been much movement in the market. Is there anything out there that you think might have a significant impact, other than unpredictable weather?
CDPT Advanced Materials Inc	Aug. 11, 2011	Transition risk	464.9	Emission Reductions	This countenance truly reflects the underlying strength of our business model and supports our efforts to date in the rapid deployment of natural energy efficiency solutions to the public and utility endmarkets.
New Jersey Resources Corp	May 4, 2018	Transition risk	298.2	Clean Energy	Various factors, ranging from unprecedented regulatory support for clean energy solutions, to rising fuel and construction costs, have made the value proposition of our scalable solutions stronger and more important than ever.
Magnentk Inc.	May 9, 2012	Transition risk	267.5	Renewable Energy	
Lime Energy Co	Aug. 12, 2009	Transition risk	267.2	Energy Efficiency	
East X North America Inc	Aug. 7, 2008	Transition risk	256.7	Clean Energy	

The table presents the excerpts in the earnings call transcripts with the highest acute, chronic and transition climate risks, respectively. The values of climate risk measures are ranked before visualization. For acute and chronic climate risks, we report the corresponding climate-related keywords and risk synonyms. For transition climate risk, we report only the climate-related keywords.

Table 3: Table 3: Transcript excerpts with highest climate risks

Table 4
Validating firm's climate risk measures

A. Correlations between physical risk measures and natural disaster data					
Dep var	Acute Risk _{<i>i,t+1</i>}		Chronic Risk _{<i>i,t+1</i>}		
	(1)	(2)	(3)	(4)	
Natural Disaster _{<i>c,t</i>}	0.0849*** (4.353)	0.0851*** (4.374)	0.0353*** (2.754)	0.0363*** (2.876)	
Natural Disaster _{<i>c,t-1</i>}		0.0041 (0.354)		-0.0038 (-0.287)	
Natural Disaster _{<i>c,t-2</i>}		-0.0145 (-1.326)		-0.0170 (-1.600)	
Natural Disaster _{<i>c,t-3</i>}		0.0045 (0.384)		0.0004 (0.028)	
Firm Attributes _{<i>i,t-1</i>}	Yes	Yes	Yes	Yes	
Industry × Time	Yes	Yes	Yes	Yes	
N	133,434	133,434	133,434	133,434	
Adj. R ²	.020	.021	.043	.052	
B. Correlations between transition risk measures and MSCI CCI					
Dep Var	Transition Risk _{<i>i,t</i>}			Nonproactive (3)	
	All (1)	Proactive (2)			
MSCI CCI _{<i>i,t</i>}	0.0512*** (3.461)	0.0446*** (3.062)	0.0468*** (3.154)		
Firm Attributes _{<i>i,t-1</i>}	Yes	Yes	Yes	Yes	
Industry × Time	Yes	Yes	Yes	Yes	
N	15,747	15,747	15,747	15,747	
Adj. R ²	.268	.142	.262		
C. Correlations between transition risk and CO ₂ intensity					
Dep Var	CO ₂ Intensity _{<i>i,t+h</i>}				
	<i>h</i> = 1	<i>h</i> = 2	<i>h</i> = 3	<i>h</i> = 4	<i>h</i> = 5
Transition Risk _{<i>i,t</i>}	Specification (1)				
	0.4531** (2.033)	0.5363** (2.104)	0.4671** (2.639)	0.5420*** (3.164)	0.6939*** (3.416)
N	2,529	2,422	2,312	2,202	2,095
Adj. R ²	.174	.245	.0944	.161	.178
Transition Risk/Nonproactive _{<i>i,t</i>}	Specification (2)				
	0.3061 (1.662)	0.3579* (1.852)	0.4082*** (3.563)	0.4449*** (2.849)	0.6449*** (4.186)
Transition Risk/Proactive _{<i>i,t</i>}	0.1758 (1.497)	0.2188 (1.403)	0.0689 (0.431)	0.1210 (0.667)	0.0609 (0.393)
N	2,529	2,422	2,312	2,202	2,095
Adj. R ²	.174	.180	.0939	.0779	.178
F-test	0.1303	0.1457	0.3393*	0.3239*	0.584***
Firm Attributes _{<i>i,t-1</i>}	Yes	Yes	Yes	Yes	Yes
Industry × Time	Yes	Yes	Yes	Yes	Yes

This table reports the validation results of our firm-level climate risk measures. In panel A, we regress the acute and chronic climate risk measures (standardized) on the occurrence of natural disasters in lagged periods. *Natural disaster* is a dummy variable that equals one if there is a natural disaster in the county where a firm was headquartered in a given quarter, zero otherwise. Columns 1 and 2 use the acute climate risk as the dependent variable, and columns 3 and 4 use the chronic climate risk as the dependent variable. Firm-level control variables (i.e., Firm attributes) include *log(Asset)*, *CapEx*, *PPE*, *Book Leverage*, *log(No. analysts)*, *Institution %*, and *Institution HHI*, all lagged by one quarter. In panel B, we regress transition risk measures on MSCI CCI. Column

Table 4: Table 4: Validation of firm climate risk measures

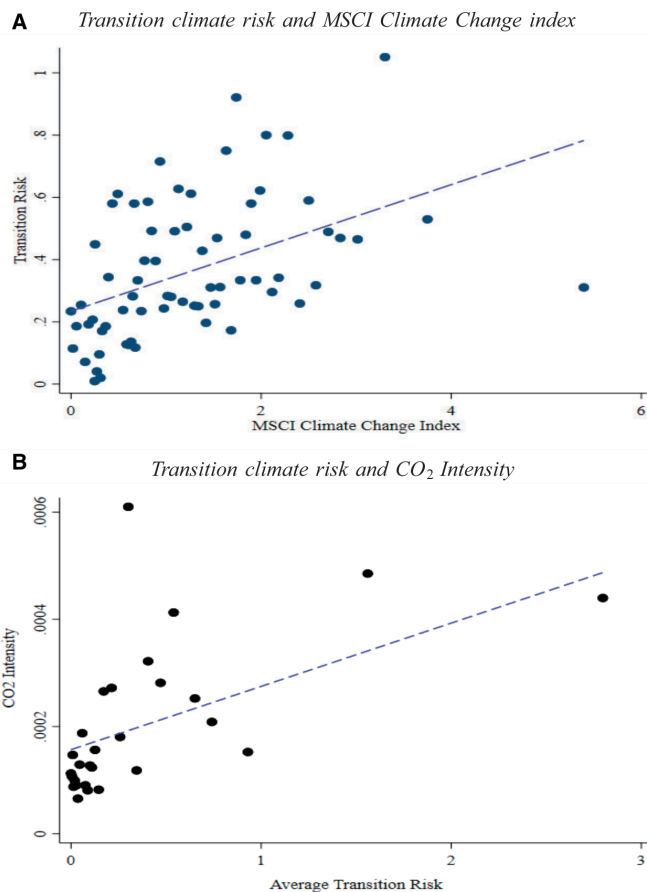


Figure 3
Scatterplots of transition climate risk and external measures
 The panels describe the correlation between the transition climate risk and two external measures. Panel A presents the (binned) scatterplot between transition climate risk and MSCI CCI for firms that have both measures available. Panel B illustrates the (binned) scatterplot of the average transition climate risk and the direct CO₂.

Figure 4: Figure 3: Transition risk versus MSCI CCI and CO₂ intensity

5.4 Table 5 and Table 6: Characteristics and pricing of transition risk

Read:

- Table 5 shows that fixed effects explain much of the variation in transition risk, but a large residual remains. This means the measure captures firm-specific variation beyond industry and time patterns.
- In Table 5, transition risk is positively related to firm size and PPE, and negatively related to book leverage. Proactive transition risk is also related to size.
- Table 6 shows a robust negative relation between transition risk and Tobin's q across horizons: the coefficients in panel A are about -0.039 to -0.042.
- When transition risk is decomposed, the nonproactive component is negative and significant, while the proactive component is small and insignificant.
- Panel B of Table 6 shows that the valuation discount is concentrated after 2010. The coefficient is about -0.057 for 2010 and later, but close to zero before 2010.

Economic message: Equity markets appear to discount transition risk only when firms do not appear to manage that risk proactively. The post-2010 result is consistent with greater investor attention and regulatory salience.

Table 5
Characteristics of climate risk measures

Model specification	A. Variance decomposition				
	Dep Var				
	Acute Risk _{i,t}		Chronic Risk _{i,t}		Transition Climate Risk _{i,t}
	Adj. R ²	Δ	Adj. R ²	Δ	Adj. R ²
Time	.009		.001		.005
Time + State	.015	.015	.008	.008	.018
Time + County	.025	.025	.040	.040	.073
Time + NAICS2	.016	.016	.030	.030	.118
Time + NAICS3	.026	.026	.043	.043	.161
Time + NAICS4	.028	.028	.075	.075	.199
Time + State + NAICS2	.020	.020	.034	.034	.124
State + NAICS2 × Time	.028	.012	.042	.012	.136
State × Time + NAICS2	.037	.021	.037	.007	.118
State × Time + NAICS2 × Time	.042	.026	.045	.015	.130
County × Time + NAICS2 × Time	.063	.047	.064	.034	.121
Firm + Time	.080	.064	.200	.170	.655
Firm + Time + Firm Attributes	.080	.064	.200	.170	.655
Firm + Time + Firm Attributes + NAICS2 × Time	.088	.072	.209	.179	.673
Firm + Time + Firm Attributes + State × Time	.097	.081	.209	.179	.657
Residual	.903		.791		.343

Dep Var	B. Firm characteristics of climate risk measures			
	Physical Risk _{i,t}		Transition Risk _{i,t}	
	Acute (1)	Chronic (2)	All (3)	Proactive (4)
log(Asset) _{i,t-1}	0.0074** (2.147)	0.0055 (0.992)	0.0138** (1.982)	0.0104*** (2.989)
CapEx _{i,t-1}	-0.0011 (-0.845)	-0.0025 (-1.184)	-0.0008 (-0.314)	0.0007 (0.480)
PPE _{i,t-1}	0.1204*** (4.687)	0.1410*** (2.907)	0.2768*** (2.773)	0.0943** (1.976)
Book Leverage _{i,t-1}	-0.0095 (-0.463)	0.0194 (0.578)	-0.1163*** (-3.318)	-0.0328* (-1.819)
log(No_Analysts) _{i,t-1}	-0.0094 (-1.463)	-0.0455*** (-3.414)	-0.0135 (-0.854)	-0.0218*** (-3.190)
Institution% _{i,t-1}	0.0304* (1.680)	-0.0028 (-0.067)	-0.0767 (-1.132)	-0.0122 (-0.498)
Institution HHI _{i,t-1}	0.0133 (0.444)	-0.0724 (-1.240)	0.0413 (0.430)	0.0239 (0.553)
Transition Risk _{i,t}				0.5858*** (11.711)
Industry × Time FE	Yes	Yes	Yes	Yes
N	124,682	124,682	124,682	124,682
Adj. R ²	.0243	.0419	.129	.386

Panel A reports the results on the adjusted R^2 from a projection of $ClimateRisk_{i,t}$ on various sets of fixed effects. Column 1 reports the adjusted R^2 of the regressions with acute climate risk as the dependent variable and different sets of fixed effects as the independent variables. In column 2, we report the change/improvement in adjusted R^2 relative to a benchmark. The benchmark for regressions in the first block is zero (no fixed effects). The benchmark for regressions in the second and third blocks is the fourth row in the first block (Time + NAICS2).

Table 6
Pricing of climate risk

Dep Var	A. All years					
	Tobin's q _{i,t+h}					
	h=1 (1)	h=3 (2)	h=5 (3)	h=1 (4)	h=3 (5)	h=5 (6)
Transition risk _{i,t}	-0.0389*** (-3.828)	-0.0404*** (-3.978)	-0.0418*** (-4.179)			
Transition risk/Nonproactive _{i,t}				-0.0416*** (-4.764)	-0.0407*** (-4.466)	-0.0405*** (-4.719)
Transition risk/Proactive _{i,t}				0.0047 (0.618)	0.0005 (0.081)	-0.0024 (-0.326)
Energy Price Exposure _{i,t}	-0.0634*** (-5.814)	-0.0577*** (-5.382)	-0.0547*** (-5.059)	-0.0601*** (-5.503)	-0.0545*** (-5.077)	-0.0517*** (-4.784)
Action Index _{i,t}				-0.0583*** (-4.458)	-0.0520*** (-3.941)	-0.0462*** (-3.455)
Firm attributes _{i,t-1}	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	111,691	104,442	97,470	111,691	104,442	97,470
Adj. R ²	.182	.210	.171	.218	.211	.210
F-test				-0.0463***	-0.0412***	-0.0381***

Dep var	B. Transition risk by different periods			
	Tobin's q _{i,t+1}			
	Year ≤ 2009 (1)	Year ≥ 2010 (2)	Year ≤ 2009 (3)	Year ≥ 2010 (4)
Transition Risk _{i,t}	-0.0041 (-0.305)	-0.0571*** (-4.911)		
Transition Risk/Nonproactive _{i,t}			-0.0151 (-1.412)	-0.0548*** (-5.234)
Transition Risk/Proactive _{i,t}			0.0174 (1.461)	-0.0045 (-0.537)
Energy Price Exposure _{i,t}	-0.0554*** (-3.729)	-0.0742*** (-5.920)	-0.0527*** (-3.546)	-0.0706*** (-5.607)
Action Index _{i,t}			-0.0426*** (-3.013)	-0.0702*** (-3.883)
Firm Attributes _{i,t-1}	Yes	Yes	Yes	Yes
Industry × Time FE	Yes	Yes	Yes	Yes
N	50,706	60,985	50,706	60,985
Adj. R ²	.180	.183	.181	.185

This table presents results from firm level regressions testing the relation between our transition climate risk measures (standardized) and Tobin's q. Panel A reports the results from regression analysis of firm's Tobin's q in different lead time periods ($t+1$, $t+3$, and $t+5$) on the lagged transition climate risk (in quarter t). In columns 1–3, the key explanatory variable is the overall transition risk measure. In columns 4–6, we decompose the transition risk measure into proactive and nonproactive components and add *Action Index* as an additional control variable. In panel B, we separately examine the relationship between Tobin's q and lagged transition climate risk in two subsample periods: 2002–2009 and 2010–2018. In both panels, all specifications include time-varying firm-level control variables, including lagged (i.e., $t-1$) *log(Asset)*, *CapEx*, *PPE*, *Book Leverage*, and *ROA* (%). Industry

Table 5: Table 5: Characteristics of climate risk measures

Table 6: Table 6: Pricing of transition climate risk

5.5 Table 7: Horse races with alternative transition risk measures

Read:

- Panel A compares the earnings-call measure with SEC filing measures from MD&A and Risk Factors sections.
- Panel B compares the earnings-call measure with a transition-risk measure constructed from firm-related news.
- Panel C compares the earnings-call measure with MSCI CCI.
- Panel D compares it with Sautner et al.-style climate exposure measures.
- Across these panels, the earnings-call transition risk measure and especially the nonproactive component remain negative and statistically significant.

Economic message: The earnings-call measure contains information that is not fully captured by filings, news, ESG scores, or alternative climate dictionaries. This is important because many climate-risk measures have limited coverage or rely on different disclosure channels.

Dep Var	Tobin's q_{t+h}							
	$h=1$ (1)	$h=5$ (2)	$h=1$ (3)	$h=5$ (4)	$h=1$ (5)	$h=5$ (6)	$h=1$ (7)	$h=5$ (8)
Transition Risk _{t}	-0.0373*** (-3.524)	-0.0425*** (-4.141)	-0.0351*** (-3.181)	-0.0422*** (-4.035)				
Transition Risk/Nonproactive _{t}					-0.0384*** (-4.126)	-0.0398*** (-4.414)	-0.0370*** (-3.828)	-0.0388*** (-4.221)
Transition Risk/Proactive _{t}					0.0019 (0.215)	-0.0047 (-0.579)	0.0033 (0.342)	-0.0062 (-0.721)
Transition Risk MDA _{t}	-0.0002 (-0.491)	-0.0062 (-0.206)			-0.0161** (-2.553)	-0.0111 (-1.390)		
Transition Risk BP _{t}					-0.0103*** (-3.185)	-0.0103*** (-4.402)	-0.0157** (-2.502)	-0.0106 (-1.557)
Energy Price Exposure _{t}	-0.0092*** (-5.094)	-0.0111*** (-4.661)	-0.0082*** (-3.036)	-0.0084*** (-4.236)	-0.0103*** (-3.185)	-0.0103*** (-4.402)	-0.0149*** (-4.729)	-0.0067*** (-3.596)
Action Index _{t}					-0.0560*** (-4.060)	-0.0427*** (-3.073)	-0.0614*** (-4.423)	-0.0447*** (-3.114)
Firm Attributes _{$t-1$}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	89,308	79,141	72,095	62,792	89,308	79,141	72,095	62,792
Adj. R ²	.186	.176	.188	.183	.187	.177	.190	.184
F-test					-0.0403*** (-3.073)	-0.0351*** (-2.855)	-0.0426*** (-3.114)	

(Continued)

Table 7: Table 7A: Horse race with SEC filing measures

Dep Var	Tobin's q_{t+h}							
	$h=1$		$h=5$		$h=1$		$h=5$	
Sample	Yes	No	Yes	No	Yes	No	Yes	No
Coverage	13%	87%	13%	87%	13%	87%	13%	87%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Transition Risk _{t}	-0.0505*** (-3.501)	-0.0325*** (-2.991)	-0.0445*** (-2.680)	-0.0377*** (-3.642)				
Transition Risk/Nonproactive _{t}					-0.0018*** (-3.773)	-0.0146*** (-3.919)	-0.0241** (-2.463)	-0.0366*** (-4.312)
Transition Risk/Proactive _{t}					0.0031 (0.999)	0.0031 (0.387)	-0.0100 (-0.869)	-0.0021 (-0.265)
MSCI CCI _{t}	-0.1703*** (-3.066)		-0.1706*** (-3.015)		-0.1661*** (-3.011)	-0.1670*** (-3.011)	-0.1696*** (-3.095)	-0.1689*** (-3.095)
Energy Price Exposure _{t}	-0.0064*** (-2.182)	-0.0601*** (-5.535)	-0.0039* (-1.912)	-0.0536*** (-4.466)	-0.0422*** (-2.900)	-0.0422*** (-2.900)	-0.0422*** (-2.900)	-0.0422*** (-2.900)
Action Index _{t}					-0.0453 (-1.171)	-0.0453 (-1.171)	-0.0453 (-1.171)	-0.0453 (-1.171)
Firm Attributes _{$t-1$}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	12,564	97,814	10,614	86,561	12,564	97,814	10,614	86,561
Adj. R ²	.212	.182	.203	.172	.215	.183	.203	.172
F-test					-0.0769*** (-3.073)	-0.0377*** (-2.855)	-0.0345*** (-2.744)	

(Continued)

Table 9: Table 7C: Horse race with MSCI CCI

Dep Var	Tobin's q_{t+h}							
	$h=1$	$h=5$	$h=1$	$h=5$	$h=1$	$h=5$	$h=1$	$h=5$
Climate News Restriction	Relevance ≥ 75							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Transition Risk _{t}	-0.0370*** (-3.405)	-0.0389*** (-3.828)			-0.0425*** (-3.937)	-0.0446*** (-4.201)		
Transition Risk/Nonproactive _{t}			-0.0398*** (-4.165)	-0.0375*** (-4.040)			-0.0432*** (-4.772)	-0.0432*** (-4.780)
Transition Risk/Proactive _{t}			0.0036 (0.601)	-0.0026 (-0.349)	0.0026 (0.411)	0.0047 (0.821)	-0.0024 (-0.338)	-0.0024 (-0.338)
Transition Risk News _{t}	-0.0051 (-0.514)	-0.0074 (-0.770)	-0.0047 (-0.481)	-0.0074 (-0.770)	0.0094 (0.841)	0.0071 (0.641)	0.0095 (0.819)	0.0095 (0.819)
Energy Price Exposure _{t}	-0.0030*** (-3.857)	-0.0040*** (-3.056)	-0.0060*** (-5.547)	-0.0101*** (-4.781)	-0.0062*** (-5.599)	-0.0033*** (-3.218)	-0.0068*** (-5.089)	-0.0033*** (-3.218)
Action Index _{t}			-0.0563*** (-4.458)	-0.0467*** (-3.455)			-0.0594*** (-4.454)	-0.0467*** (-3.455)
Firm Attributes _{$t-1$}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	111,691	97,470	111,691	97,470	111,691	97,470	111,691	97,470
Adj. R ²	.182	.171	.183	.172	.182	.171	.183	.172
F-test			-0.0444*** (-3.073)	-0.0349*** (-2.855)			-0.0489*** (-3.114)	

(Continued)

Table 8: Table 7B: Horse race with news measures

Sample	Tobin's q_{t+h}				$h=1$	
	$h=1$	$h=5$	All Years	$h=5$	Year=2010	Year=2010
	(1)	(2)	(3)	(4)	(5)	(6)
Transition Risk _{t}	-0.0383*** (-3.451)	-0.0386*** (-3.363)			-0.0484*** (-3.854)	-0.0475*** (-4.011)
Transition Risk/Nonproactive _{t}			-0.0413*** (-3.838)	-0.0391*** (-3.677)		-0.0475*** (-4.011)
Transition Risk/Proactive _{t}			0.0029 (0.361)	-0.0029 (-1.132)		-0.0075 (-0.822)
CCExposure _{t}	0.0145 (0.749)	0.0026 (0.128)			-0.0122 (-0.629)	
CCExposure _{t} ^{Pro}			-0.0027 (-0.252)	-0.0039 (-0.176)		0.0065 (0.422)
CCExposure _{t} ^{Opp}			0.0216 (1.574)	0.0146 (1.094)		0.0055 (0.346)
CCExposure _{t} ^{Reg}			0.0075 (0.397)	0.0041 (0.200)		-0.0216* (-1.680)
Energy Price Exposure _{t}	-0.1022*** (-7.904)	-0.0883*** (-7.028)	-0.0979*** (-7.680)	-0.0857*** (-7.484)	-0.1159*** (-7.534)	-0.1129*** (-7.385)
Action Index _{t}			-0.0513*** (-3.820)	-0.0459*** (-3.274)		-0.0595*** (-3.352)
Firm Attributes _{$t-1$}	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	124,444	109,730	124,444	109,730	71,224	71,224
Adj. R ²	.151	.149	.152	.150	.159	.161
F-test			-0.0384*** (-3.073)	-0.0397*** (-3.114)		-0.0409*** (-3.114)

This table presents the horse-race test results when we regress Tobin's q in different lead time periods on both our transition risk measures using earnings call transcript data and other transition risk measures constructed from alternative data source. In panel A, the alternative transition risk measures are the measure based on the MDA section of SEC filings (columns 1-2, 5-6) and the measure based on the Risk Factors section of SEC filings (columns 3 and 4, 7 and 8), respectively. The alternative risk measures in Panel B are constructed from company news data from RavenPack database. *Transition risk news* is equal to the number of news articles related to the firm's transition climate risk exposure divided by the number of all news articles related to the company. In column 1 to column 4, the news articles are filtered by relevance score higher than 75. According to RavenPack, values above 75 are considered significantly relevant. Columns 5 to column 6 present the results when we change the relevance cutoff to 50. In Panel C, the alternative transition risk measure is the MSCI CCI. Column 1, 3, 5, and 7 present the regression results on the overlapped sample (13% of our sample). Column 2, 4, 6, and 8 present the results on the other part of our sample (87% of our sample) that is not covered in MSCI CCI. In panel D, we use the climate exposure measures from Sautner et al. (2013). Specifically, *CCExposure* is the relative frequency with which bigrams related to climate change occur in the transcripts of earnings calls. *CCExposure^{Pro}* is the relative frequency with which bigrams that capture physical shocks related to climate change occur in the transcripts of earnings calls. *CCExposure^{Opp}* is the relative frequency with which bigrams that capture opportunities related to climate change occur in the transcripts of earnings calls. *CCExposure^{Reg}* is the relative frequency with which bigrams that capture regulatory shocks related to climate change occur in the transcripts of earnings calls. Legend firm attributes: *Log(Average)*, *CapEx*, *PPE*, *Book Leverage*, and *ROE (%)* and industry by quarter fixed effects are included in all tests of each panel. Table A.1 in the appendix contains detailed definitions of all variables. Standard errors are double clustered at the firm and quarter levels. *t*-statistics are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Table 7D: Horse race with Sautner et al. measures

5.6 Table 8 and Table 9: Within-firm pricing and investment response

Read:

- Table 8 uses changes in transition risk and changes in Tobin's q within firms.
- In panel A, changes in transition risk predict lower future changes in Tobin's q at longer horizons, especially $h = 4$ to $h = 6$.
- In panel B, changes in nonproactive transition risk are negative and significant at longer horizons, while changes in proactive risk are not.
- Table 9 shows no significant total effect of transition risk on CapEx.
- Once transition risk is decomposed, proactive transition risk predicts higher CapEx at $h = 1$, $h = 3$, and $h = 5$.

Economic message: The within-firm result strengthens the pricing interpretation, and the investment result shows that proactive firms respond to transition risk through real capital investment.

Table 8
Pricing of within-firm climate risk

A. Total transition risk						
Dep Var	Δ Tobin's $q_{i,t+h}$					
	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	$h=5$ (5)	$h=6$ (6)
Δ Transition Risk $_{i,t}$	-0.0003 (-0.164)	-0.0025 (-1.023)	-0.0030 (-1.177)	-0.0053** (-2.066)	-0.0034* (-1.824)	-0.0046** (-2.127)
Energy Price Exposure $_{i,t}$	0.0008 (0.477)	0.0032 (1.216)	0.0037 (1.350)	0.0030 (0.887)	0.0041 (1.188)	0.0041 (1.124)
Firm Attributes $_{i,t-1}$ FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	110,761	106,830	103,113	99,421	95,929	92,554
Adj. R^2	.103	.188	.311	.301	.392	.427
B. Proactive and nonproactive transition risk						
Dep Var	Δ Tobin's $q_{i,t+h}$					
	$h=1$ (1)	$h=2$ (2)	$h=3$ (3)	$h=4$ (4)	$h=5$ (5)	$h=6$ (6)
Δ Transition Risk/ Nonproactive $_{i,t}$	-0.0009 (-0.454)	-0.0026 (-1.222)	-0.0035 (-1.571)	-0.0046* (-1.944)	-0.0033* (-1.857)	-0.0050** (-2.137)
Δ Transition Risk/Proactive	0.0010 (0.853)	-0.0002 (-0.165)	0.0006 (0.435)	-0.0012 (-0.872)	-0.0004 (-0.307)	0.0004 (0.498)
Energy Price Exposure $_{i,t}$	0.0009 (0.531)	0.0035 (1.333)	0.0041 (1.501)	0.0037 (1.104)	0.0050 (1.443)	0.0049 (1.351)
Action Index $_{i,t}$	-0.0025 (-1.220)	-0.0055* (-1.751)	-0.0080** (-2.004)	-0.0123** (-2.642)	-0.0150*** (-3.039)	-0.0155*** (-2.934)
Firm Attributes $_{i,t-1}$	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	110,761	106,830	103,113	99,421	95,929	92,554
Adj. R^2	.103	.188	.312	.301	.392	.428

This table presents the results from firm level regressions testing the relation between the change in transition climate risk measures (standardized) and the change in Tobin's q while controlling for firm fixed effects. Panel A reports the results from regression analysis of change in Tobin's q in different lead time periods ($t+1$ to $t+6$) on the lagged change in transition climate risk. The key explanatory variable is the change in transition risk measure from $t-1$ to t . In panel B, we decompose the change in transition risk measure into the change in proactive and nonproactive components and add *Action Index* as an additional control variable. In both panels, all specifications include time-varying firm-level control variables, including lagged (i.e., $t-1$) Tobin's q , $\log(\text{Asset})$, CapEx , PPE , Book Leverage , and ROA (%). Industry (NAICS three-digit) by quarter fixed effects are also included in all tests. We exclude the firms in finance and insurance sector. Table A.1 in the appendix

Table 11: Table 8: Pricing of within-firm climate risk

Table 9
Predicting the firm's investment

Dep Var	$\text{CapEx}_{i,t+h}$					
	$h=1$ (1)	$h=3$ (2)	$h=5$ (3)	$h=1$ (4)	$h=3$ (5)	$h=5$ (6)
Transition Risk $_{i,t}$	0.0480 (1.460)	0.0498 (1.499)	0.0421 (1.256)			
Transition Risk/Nonproactive $_{i,t}$				0.0236 (0.783)	0.0158 (0.547)	0.0074 (0.252)
Transition Risk/Proactive $_{i,t}$				0.0460*** (2.951)	0.0623*** (3.455)	0.0641*** (3.287)
Energy Price Exposure $_{i,t}$	0.0896* (1.901)	0.1120** (2.435)	0.1186** (2.557)	0.0867* (1.836)	0.1030** (2.236)	0.1136** (2.452)
Action Index $_{i,t}$				0.0058 (0.274)	0.0782*** (3.442)	0.0201 (0.909)
Firm Attributes $_{i,t-1}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	126,099	118,043	110,313	126,099	118,043	110,313
Adj. R^2	.439	.437	.435	.439	.438	.435
F-test				-0.0224	-0.0465	-0.0567*

This table reports estimates of the regressions of capital expenditures (in different lead time periods) on transition risk. Columns 1–3 shows the results using *Transition risk* as the key explanatory variable. In columns 4–6, we replace transition risk measure with its two components: nonproactive and proactive transition risk, and we add *Action index* as additional control variable. Lagged $\log(\text{Asset})$ and industry by quarter fixed effects are included in all tests. Table A.1 in the appendix defines all the variables. Standard errors are double clustered at the firm and quarter levels. t -statistics are shown in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

in $t+3$ and $t+5$.³² The estimates are economically meaningful, representing approximately 1.6%–2.3% of the average investment level. In the bottom

Table 12: Table 9: Predicting capital expenditures

5.7 Table 10: R&D, green innovation, and employment

Read:

- Panel A shows that total transition risk predicts lower future R&D spending. The negative relation is driven by the nonproactive component.
- Panel B shows that transition risk is associated with green patenting, especially among firms with patents. The proactive component is positively related to green-patent output.
- Panel C shows that total transition risk predicts lower employment, again driven by nonproactive transition risk. Proactive transition risk is essentially zero in the employment regressions.
- The pattern across outcomes is consistent: nonproactive transition risk is associated with valuation discounts, reduced R&D, and reduced employment, while proactive risk is associated with investment and green innovation.

Economic message: Firms facing climate transition risk do not all react in the same way. The market and real outcomes distinguish between passive exposure and active adaptation.

Table 10
Predicting the firm's other responses

Dep Var	A. R&D expenditures					
	R&D investment _{it,t+1}			R&D investment _{it,t+1}		
	h=1	h=2	h=3	h=1	h=2	h=3
	(1)	(2)	(3)	(4)	(5)	(6)
Transition Risk _{it,t}	-0.055** (-2.393)	-0.052** (-2.269)	-0.050** (-2.391)	-0.048** (-2.007)	-0.050** (-2.241)	-0.055** (-2.478)
Transition Risk/Nonproductive _{it,t}				-0.003 (-0.01)	-0.002 (-0.026)	-0.001 (-0.101)
Energy Price Exposure _{it,t}	-0.179*** (-7.917)	-0.178*** (-7.758)	-0.174*** (-7.701)	-0.163*** (-7.174)	-0.162*** (-7.107)	-0.156*** (-7.101)
Action Index _{it,t}				-0.244*** (-11.879)	-0.245*** (-10.868)	-0.245*** (-10.758)
Firm Attributes _{it,t-1}	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	128,303	119,997	111,971	128,303	119,997	111,971
Adj. R ²	.388	.381	.373	.398	.389	.382
F-test				-0.0515**	-0.0578**	-0.0532**

Dep Var	B. Green patents (annual)				Green patents ratio _{it,t+1}			
	Green patents _{it,t+1}				Green patents ratio _{it,t+1}			
	h=1	h=2	h=1	h=2	h=1	h=2	h=1	h=2
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Transition Risk _{it,t}	0.0115 (1.798)	0.0080 (1.276)	0.0087 (0.804)	0.0014 (0.200)	0.021*** (3.914)	0.0331*** (3.803)	0.0198** (2.252)	0.0168* (1.256)
Transition Risk/Nonproductive _{it,t}			0.0007 (0.006)	0.0001 (0.001)			0.0193** (2.959)	0.021*** (3.705)
Energy Price Exposure _{it,t}	0.0332*** (4.070)	0.032*** (4.003)	0.0309** (0.803)	0.0340*** (4.700)	0.0207** (2.525)	0.0231** (2.367)	0.0188*** (2.739)	0.028*** (2.307)
Action Index _{it,t}			0.0063 (1.585)	0.0026 (0.542)			-0.0019 (-0.028)	-0.0018 (-0.088)
Firm Attributes _{it,t-1}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry × Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	32,713	32,713	32,713	32,713	8,372	8,106	8,372	8,106
Adj. R ²	.199	.193	.192	.193	.103	.110	.109	.122
F-test			-0.0013	-0.0009			-0.0004	-0.0004

(Continued)

Table 13: Table 10A-B: R&D and green patents

Dep Var	C. Employment (annual)			
	log(employment) _{it,t+1}			
	h=1	h=2	h=1	h=2
	(1)	(2)	(3)	(4)
Transition Risk _{it,t}	-0.0195** (-2.050)	-0.0202** (-2.047)		
Transition risk/nonproductive _{it,t}			-0.0188* (-1.711)	-0.0197* (-1.692)
Transition risk/productive _{it,t}			-0.0000 (-0.003)	0.0002 (0.022)
Energy price exposure _{it,t}	0.0032 (0.249)	0.0007 (0.050)	-0.0001 (-0.325)	-0.0067 (-0.510)
Action Index _{it,t}			0.0634*** (6.647)	0.0624*** (6.267)
Firm attributes _{it,t-1}	Yes	Yes	Yes	Yes
Industry × Time FE	Yes	Yes	Yes	Yes
N	32,165	30,533	32,165	30,533
Adj. R ²	.776	.771	.778	.771
F-test			-0.0188	-0.0199

In panel A, we regress *R&D Investment* (in $t+1$, $t+2$, $t+3$) on overall transition risk measure (in columns 1-3) and decomposed transition risk measures (in columns 4-6), respectively. In columns 1-4 of panel B, the dependent variable is *Green patents ratio*, a dummy variable equals one if a firm has at least one green patent, and zero otherwise. The sample includes all firms. In columns 5-8 of panel B, the dependent variable is *Green patents ratio*, the number of green patents scaled by the total number of patents in the year. The sample is restricted to the firms with patents. In panel C, the dependent variable is the natural logarithm of the firm's employment level. All specifications include lags (i.e., $t-1$) *log(Asize)* as the control variable. Industry by quarter fixed effects are included in all tests. Table A.1 in the appendix defines all the variables. Standard errors are double clustered at the firm and quarter levels. t -statistics are shown in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 14: Table 10C: Employment response

5.8 Appendix Table A.1: variable definitions

Read:

- The appendix table records definitions and data sources for climate measures, firm attributes, external validation measures, and response variables.
- It is useful as a reference when interpreting the regressions because many controls are lagged, standardized, or measured at different frequencies.

Appendix

Table A.1
Variable definitions

Variable name	Description	Source
<i>Acute climate risk</i>	The frequency of mentions of the unigrams or bigrams related to the acute climate discussion in the proximity of risk synonyms, divided by the total length of the transcript, and then multiplied by 10^4	StreetEvents
<i>Chronic climate risk</i>	The frequency of mentions of the unigrams or bigrams related to the chronic climate discussion in the proximity of risk synonyms, divided by the total length of the transcript, and then multiplied by 10^4	StreetEvents
<i>Transition climate risk</i>	The frequency of mentions of the unigrams or bigrams related to the transition climate discussion, scaled by the total length of the transcript, and then multiplied by 10^4	StreetEvents
<i>Transition risk/proactive</i>	The frequency of mentions of the unigrams or bigrams related to the transition climate discussion in the proximity of proactive verbs, divided by the total length of the transcript, and then multiplied by 10^4	StreetEvents
<i>Transition risk/nonproactive</i>	The frequency of mentions of the unigrams or bigrams related to the transition climate discussion which are not in the proximity of proactive verbs, divided by the total length of the transcript, and then multiplied by 10^4	StreetEvents
<i>Energy price exposure</i>	The number of sentences that jointly mentions synonyms of "energy" synonyms and "price" (two words not necessarily synonyms for each other), divided by the total number of sentences in the earnings call transcript. Synonyms of "energy" include gas, fuel, oil, and energy. Synonyms of "price" include cost, expense, price, costs, expenses, and prices	StreetEvents
<i>Action index</i>	The frequency of mentions of the "proactive" verbs in the entire transcript (except those near, within ± 1 sentences of, climate-related discussions), divided by the length of the transcript	StreetEvents
<i>Disaster dummy</i>	A dummy variable equal to one if there is a natural disaster in the same county where a firm was headquartered	SHELDUS
<i>CO₂ intensity</i>	Sum of CO ₂ emissions of all plants operated by the firm, scaled by the total assets	EPA
<i>Tobin's q</i>	(Total assets + Market value of equity - Book value of equity) / Total assets	Compustat
<i>CapEx</i>	Capital expenditures, scaled by the total assets of the previous quarter end	Compustat
<i>R&D</i>	Research & Development expenditures, scaled by the total assets of the previous quarter end	Compustat
<i>log(Employment) (annual)</i>	Natural logarithm of firm's employment	Compustat
<i>I(Green patents) (annual)</i>	A dummy variable that equals one if a firm has at least one green patent in the year, and zero otherwise. Green patents are identified following the OECD classification	Global Corporate Patent data set
<i>Green patent ratio (annual)</i>	The number of green patents scaled by the total number of patents in the year	Global Corporate Patent data set
<i>log(Asset)</i>	Natural logarithm of firm's total assets.	Compustat
<i>PPE</i>	Property, Plant and Equipment, scaled by total assets of the previous quarter end.	Compustat
<i>Book Leverage</i>	Total debt (= short-term debt + long-term debt), scaled by the total assets.	Compustat
<i>log(No Analysts)</i>	The natural logarithm of number of analysts covering the	I/B/E/S

Table 15: Appendix Table A.1: variable definitions, part 1

Table A.1
(Continued)

Variable name	Description	Source
Institution %	The percentage of institutional ownership.	Thomson-Reuters Institutional Holdings (I3F)
Institution HHI	The Herfindahl-Hirschman Index of institutional ownership.	Thomson-Reuters Institutional Holdings (I3F)
ROA	Operating Income Before Depreciation (OIBDPQ), scaled by total assets of the previous quarter end, multiply by 100.	Compustat
Transition Risk MDA	The transition climate risk measure based on the management discussion and analysis section of SEC filings.	10K/10Q
Transition Risk RF	The transition climate risk measure based on the risk factors section of SEC filings.	10K/10Q
Transition Risk News	The number of news articles related to the firm's transition climate risk exposure divided by the total number of news articles related to the firm.	RavenPack
MSCI Climate Change Index (CCI)	The climate change materiality weight $\times$ the climate change risk rating. The materiality weight measures the importance of climate change to a firm's financial performance. The climate change risk rating is calculated as (10 - climate change theme score). Climate change theme score is a continuous variable ranging from 0 to 10, with higher value indicating better performance (i.e., lower risk).	MSCI
RepRisk Environmental Score	The environmental component of ESG rating provided by RepRisk	RepRisk

Table 16: Appendix Table A.1: variable definitions, part 2

6 Short summary

- The paper constructs firm-level climate risk measures from earnings call transcripts.
- It separately measures acute physical risk, chronic physical risk, and transition risk.
- Physical risk discussions must occur near risk-related language; transition risk is broader and does not require risk-synonym proximity.
- The transition risk measure is decomposed into proactive and nonproactive components using action-oriented verbs near transition-risk terms.
- The sample covers 4,719 U.S. public firms and 139,959 firm-quarter observations from 2002 to mid-2018.
- The measures are validated using natural disasters, MSCI CCI, CO₂ intensity, and transcript examples.
- Transition risk is negatively related to Tobin's q , especially after 2010.
- The valuation discount is concentrated in the nonproactive component of transition risk.
- Horse-race regressions show that the earnings-call measure is not subsumed by SEC filing measures, news measures, MSCI CCI, or Sautner et al.-style measures.
- Within-firm changes in nonproactive transition risk predict declines in Tobin's q .
- Proactive firms increase CapEx when transition risk rises.
- Nonproactive transition risk is associated with lower R&D and lower employment.
- Proactive transition risk is associated with green patenting, especially among firms with patenting activity.
- The main message is that climate risk pricing and firm behavior depend on whether firms actively respond to transition risk.

7 Conclusion

7.1 Core conclusion

Li, Shan, Tang, and Yao provide a firm-level measurement framework for climate risk using earnings-call text. Their key empirical finding is that transition climate risk is priced in equity markets, but the discount is concentrated among firms that do not appear to respond proactively.

7.2 Economic intuition

Climate transition risk can be costly because regulation, technology shifts, consumer preferences, and investor attention can reduce the value of carbon-intensive or poorly adapted business models. But firms can also respond through investment, innovation, and operational change. The market therefore values not only exposure, but also the firm's apparent ability and willingness to adapt.

7.3 Takeaway

The paper's most useful insight is the distinction between exposure and response. Measuring climate risk at the firm level requires more than counting climate words. It requires separating physical from transition risk and, within transition risk, distinguishing passive discussion from proactive adaptation.